

Name: Recitation 09/01/2023

(Suha's answer! Open to any comments! Let me know.)

1. **Consideration sets:** Suppose an individual has alternatives given by $X = \{a, b, c, d, e\}$ and a domain of $\mathcal{D} = \{\{a, b\}, \{a, b, c\}, \{a, b, c, d\}, \{a, b, c, d, e\}\}$. Suppose an individual only considers a subset of alternatives from each menu. The subset of alternatives considered from each menu is given by the following mapping $\Gamma : \mathcal{D} \rightarrow X$. We have consideration sets given by

M	$\Gamma(M)$
$\{a, b\}$	$\{a\}$
$\{a, b, c\}$	$\{b, c\}$
$\{a, b, c, d\}$	$\{c, d\}$
$\{a, b, c, d, e\}$	$\{a, b, c, d, e\}$

and transitive and complete preferences given by the order $e \succ d \succ c \succ b \succ a$. Suppose one individual chooses according to $C_{\succ}(M) = \{x \mid x \succ y \ \forall y \in M \setminus \{x\}\}$ and $C_{\Gamma, \succ} = \{x \mid x \succ y \ \forall y \in \Gamma(M) \setminus \{x\}\}$.

- (a) Compute the choices from each menu in the domain according to $C_{\succ}$. *"I think this 'a' comes from $T(M)$ (not M) as it is excluded from $T(M)$ in the first place when considering 'y'."*

M	$C_{\succ}(M)$
$\{a, b\}$	b
$\{a, b, c\}$	c
$\{a, b, c, d\}$	d
$\{a, b, c, d, e\}$	e

- (b) Compute all the choices from each menu in the domain according to $C_{\Gamma, \succ}$.

M	$C_{\Gamma, \succ}$
$\{a, b\}$	a
$\{a, b, c\}$	c
$\{a, b, c, d\}$	d
$\{a, b, c, d, e\}$	e

- (c) Suppose that you have observed the choices from the two choice functions and generate a revealed preference where $x R y$ when for some menu M we have $x \in C(M)$ and $y \in M$. How do the two revealed preference relations compare?

For $C_{\succ}(M)$:

R	a	b	c	d	e
a	?	b	c	d	e
b		b	c	d	e
c			c	d	e
d				d	e
e					e

For $C_{\Gamma, \succ}(M)$:

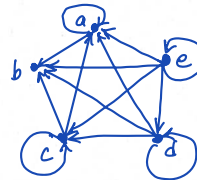
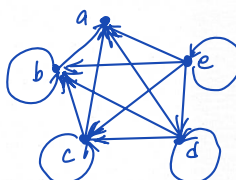
R	a	b	c	d	e
a	a	a	c	d	e
b	a	?	c	d	e
c			c	d	e
d				d	e
e					e

Both are revealed incomplete but transitive.

To see transitivity:

$$(x \rightarrow y \Leftrightarrow x > y)$$

look for "triangles" 



(d) What happens when the above holds but $\Gamma(\{a, b, c, d, e\}) = \{b, c\}$

We can reveal from $C_P, >$ that:

R	a	b	c	d	e
a	a	a	c	d	?
b	a	?	c	d	?
c			c	d	?
d			c	d	?
e			c	d	?

If we define the indifference as $x \sim y \Leftrightarrow x R y \& y R x$, then $c \sim d$ may be revealed.

2. Convexity:

(a) Suppose the function $f : \mathbb{R}^N \rightarrow \mathbb{R}$ is concave. Prove that f is quasi-concave.

pf) Let $x, y \in \mathbb{R}^N$. WLOG, we may assume $f(x) \leq f(y)$.

Let $\lambda \in [0, 1]$ be given.

(We shall show: $f(\lambda x + (1-\lambda)y) \leq f(y)$ by the def of quasi-concavity.)

Then

$$\begin{aligned} f(\lambda x + (1-\lambda)y) &\leq \lambda f(x) + (1-\lambda)f(y) \quad (\text{by the def of concavity}) \\ &\leq \lambda f(y) + (1-\lambda)f(y) \quad (\because f(x) \leq f(y)) \\ &= f(y) \end{aligned}$$

Thus, f is quasi-concave.

Graded!

(b) Suppose $f : \mathbb{R}^N \rightarrow \mathbb{R}$ is concave and $g : \mathbb{R} \rightarrow \mathbb{R}$ is monotonic increasing. Is $g \circ f$ concave? If so, prove it. If not, find a counter example and explain why.

False.

(Counter-example)

$$f(x) = \ln x^2$$

$$g(x) = e^x$$

$$\text{Then } g \circ f(x) = e^{\ln x^2} = x^2 : \text{ a convex function.}$$

$$\delta_C: C \rightarrow \mathbb{R} \quad \delta_C: \mathbb{R}^N \rightarrow \mathbb{R}$$

(c) Consider the function

$$\delta_C(y) = \begin{cases} 0 & \text{if } y \in C \\ -\infty & \text{otherwise} \end{cases}$$

where $C \subset \mathbb{R}^N$. Can you come up with a condition on C so the function is convex? Prove C is convex for the case you study.

Prop If a function is concave/convex,
the function is continuous.

$\Leftrightarrow$ If Not continuous, not convex! $\rightarrow$ e.g.,

To have $\delta_C(y)$ continuous,

$C = \emptyset$ or $\mathbb{R}^N$.

